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Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 - ASSESSMENT TOOL 2

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO1: Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method.

(BL-1, BL-2, BL-3)

Assessment Tool Weightage:

QUESTIONS: 10 **Total Marks: 100**

Annexure A

Concept Mapping

Code of Conduct:

I S.Chandra Siddhardha(192525222) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Chandra Siddhardha

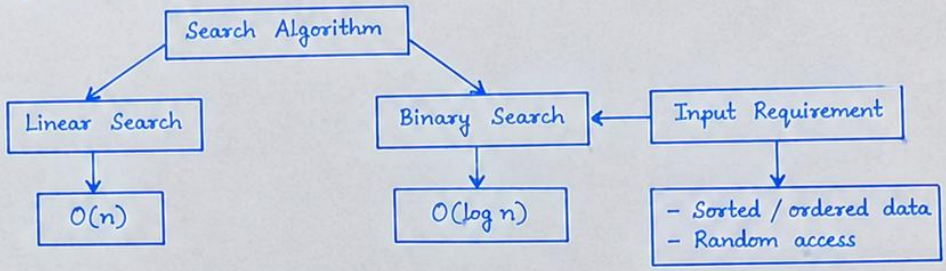
RUBRICS FOR CALCULATION

Criteria	**Weigh tage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

10. Consider the concept map: Search Algorithm → Linear Search → O(n) and Search Algorithm → Binary Search → O(log n). Construct the concept map with two parallel branches and explain how each searching approach leads to a different complexity result. Extend the map by adding Input Requirement as a node connected to Binary Search. Justify why binary search is more efficient than linear search for large input sizes. Interpret how this map helps in comparing the searching algorithms.

Answer:

The completed concept map is shown below:



1. **Linear Search → O(n):**
   - Linear search checks each element one by one until the target is found or the list ends.
   - Hence, the time complexity is O(n), where n is the number of elements.

2. **Binary Search → O(log n):**
   - Binary search divides the search space into half at each step.
   - It requires the data to be sorted and supports random access.
   - Therefore, the time complexity is O(log n).

3. **Input Requirement (for Binary Search):**
   - Binary search works only on sorted / ordered data.
   - It also requires random access to the data (e.g., arrays).

4. **Why Binary Search is More Efficient for Large Inputs:**
   - O(log n) grows much slower than O(n) as n increases.
   - For example, when n = 1,000,000, linear search may require up to 1,000,000 comparisons, while binary search needs only about 20 comparisons ($\log_2 10^6 \approx 20$).
   - Thus, binary search saves a lot of time for large input sizes.

5. **Interpretation:**
   - This concept map shows two different searching approaches and their corresponding complexities.
   - It helps in comparing the algorithms by highlighting the effect of input conditions (sorted data) and showing why binary search is preferred for large datasets.

Conclusion:

The concept map clearly shows that linear search is simple but less efficient, while binary search is faster for large inputs, provided the input is sorted. This makes it easy to compare and choose the right searching algorithm based on the problem constraints.